

## CHYULBYU, Russian Federation

WMO: 311230

Lat: 57.816N

Lon: 130.900E

Elev: 836

StdP: 14.26

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |              |            |              | MCWS/PCWD<br>to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|--------------|------------|--------------|--------------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |              | 1%         |              |                          |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB         | WS         | MCDB         | MCWS                     | PCWD       |              |
| (a)<br>1      | (b)<br>-50.0 | (c)<br>-47.0 | (d)<br>-53.4                  | (e)<br>0.2 | (f)<br>-49.4 | (g)<br>-50.5 | (h)<br>0.3 | (i)<br>-46.8 | (j)<br>9.5            | (k)<br>-17.9 | (l)<br>7.8 | (m)<br>-17.6 | (n)<br>0.4               | (o)<br>170 | (p)<br>0.665 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 25.1                   | 86.2            | 66.5 | 82.0 | 64.7 | 78.0 | 63.4 | 69.4                | 80.1 | 67.4 | 77.8 | 65.4 | 75.1 | 2.8                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 65.7                            | 97.8 | 73.9 | 63.8 | 91.5 | 72.1 | 61.7 | 84.7 | 69.6 | 34.1          | 80.3 | 32.4 | 78.2 | 30.8 | 75.2 | 84.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |           |                    |         | n-Year Return Period Values of Extreme Temperature |           |            |           |            |           |            |           |          |  |
|-------------------|---------|---------|----------------------------|-----------|--------------------|---------|----------------------------------------------------|-----------|------------|-----------|------------|-----------|------------|-----------|----------|--|
|                   |         |         | Mean                       |           | Standard Deviation |         | n=5 years                                          |           | n=10 years |           | n=20 years |           | n=50 years |           |          |  |
| 1%                | 2.5%    | 5%      | Min                        | Max       | Min                | Max     | Min                                                | Max       | Min        | Max       | Min        | Max       | Min        | Max       |          |  |
| (a) 7.4           | (b) 6.0 | (c) 5.1 | (e) -52.3                  | (f) 92.4  | (g) 3.2            | (h) 3.5 | (i) -54.7                                          | (j) 94.9  | (k) -56.6  | (l) 97.0  | (m) -58.4  | (n) 98.9  | (o) -60.7  | (p) 101.5 |          |  |
|                   |         |         | DB                         |           |                    |         |                                                    |           |            |           |            |           |            |           |          |  |
|                   |         |         | WB                         |           |                    |         |                                                    |           |            |           |            |           |            |           |          |  |
|                   |         |         |                            | (e) -52.3 | (f) 74.4           | (g) 3.2 | (h) 2.1                                            | (i) -54.6 | (j) 75.8   | (k) -56.5 | (l) 77.0   | (m) -58.3 | (n) 78.2   | (o) -60.6 | (p) 79.7 |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 20.0   | -29.3 | -19.8 | 2.8   | 26.7  | 45.9  | 59.1  | 64.4  | 59.0  | 44.2  | 22.7  | -9.3  | -29.0 |  |  |
|                                                                        | DBStd       | 35.34  | 9.92  | 9.95  | 11.82 | 10.26 | 6.94  | 6.31  | 5.66  | 6.60  | 7.46  | 10.86 | 13.85 | 10.23 |  |  |
|                                                                        | HDD50       | 12039  | 2459  | 1956  | 1464  | 700   | 163   | 8     | 0     | 10    | 205   | 847   | 1778  | 2449  |  |  |
|                                                                        | HDD65       | 16527  | 2924  | 2376  | 1929  | 1150  | 591   | 193   | 81    | 204   | 625   | 1312  | 2228  | 2914  |  |  |
|                                                                        | CDD50       | 1083   | 0     | 0     | 0     | 0     | 37    | 280   | 446   | 290   | 30    | 0     | 0     | 0     |  |  |
|                                                                        | CDD65       | 98     | 0     | 0     | 0     | 0     | 0     | 15    | 63    | 20    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 1796   | 0     | 0     | 0     | 0     | 40    | 531   | 885   | 338   | 2     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 558    | 0     | 0     | 0     | 0     | 7     | 154   | 311   | 86    | 0     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 1.7    | 1.5   | 1.3   | 1.7   | 2.1   | 2.1   | 1.8   | 1.6   | 1.5   | 1.8   | 2.0   | 1.8   | 1.5   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 15.4   | 0.4   | 0.2   | 0.3   | 0.5   | 1.6   | 2.2   | 2.8   | 3.0   | 2.2   | 1.0   | 0.6   | 0.4   |  |  |
|                                                                        | PrecMax     | 23.6   | 1.0   | 0.6   | 0.7   | 1.3   | 3.1   | 4.3   | 5.3   | 6.1   | 5.4   | 3.7   | 2.3   | 0.9   |  |  |
|                                                                        | PrecMin     | 10.7   | 0.2   | 0.0   | 0.1   | 0.1   | 0.5   | 0.9   | 0.6   | 0.5   | 0.8   | 0.2   | 0.2   | 0.2   |  |  |
|                                                                        | PrecStd     | 2.9    | 0.2   | 0.1   | 0.2   | 0.4   | 0.7   | 0.9   | 1.2   | 1.3   | 1.0   | 0.7   | 0.4   | 0.2   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | -0.8  | 13.2  | 41.9  | 62.3  | 79.0  | 89.6  | 91.8  | 87.5  | 71.8  | 50.7  | 30.1  | 4.3   |  |  |
|                                                                        |             | MCWB   | -1.3  | 11.0  | 33.6  | 46.0  | 57.1  | 65.8  | 68.2  | 68.8  | 58.9  | 42.0  | 27.4  | 3.6   |  |  |
|                                                                        | 2%          | DB     | -6.8  | 6.1   | 34.8  | 54.1  | 72.9  | 84.5  | 87.5  | 81.8  | 66.2  | 43.9  | 20.7  | -4.5  |  |  |
|                                                                        |             | MCWB   | -7.3  | 4.8   | 28.7  | 41.2  | 53.8  | 62.9  | 68.1  | 66.5  | 56.0  | 38.4  | 19.1  | -5.1  |  |  |
|                                                                        | 5%          | DB     | -10.6 | 0.2   | 29.6  | 48.9  | 68.2  | 80.6  | 83.3  | 77.4  | 61.5  | 40.0  | 15.5  | -9.3  |  |  |
|                                                                        |             | MCWB   | -11.1 | -0.9  | 24.7  | 38.3  | 51.9  | 61.2  | 66.7  | 64.6  | 53.1  | 35.7  | 14.3  | -9.8  |  |  |
|                                                                        | 10%         | DB     | -14.3 | -4.5  | 24.4  | 44.3  | 63.1  | 76.2  | 78.5  | 73.1  | 57.1  | 36.2  | 10.7  | -14.1 |  |  |
|                                                                        |             | MCWB   | -14.7 | -5.4  | 20.5  | 35.3  | 49.1  | 60.1  | 65.5  | 62.8  | 50.4  | 32.6  | 9.7   | -14.5 |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | -1.3  | 11.3  | 33.8  | 46.7  | 58.4  | 69.0  | 73.0  | 71.0  | 60.8  | 42.9  | 27.1  | 3.6   |  |  |
|                                                                        |             | MCDB   | -0.9  | 13.1  | 41.0  | 62.2  | 74.2  | 80.6  | 84.7  | 81.6  | 69.2  | 48.5  | 30.1  | 4.2   |  |  |
|                                                                        | 2%          | WB     | -7.3  | 4.8   | 28.9  | 41.7  | 55.0  | 66.1  | 70.5  | 68.3  | 57.1  | 39.2  | 19.4  | -5.0  |  |  |
|                                                                        |             | MCDB   | -6.8  | 6.0   | 34.5  | 52.9  | 69.0  | 79.1  | 81.9  | 78.2  | 64.0  | 43.0  | 20.8  | -4.5  |  |  |
|                                                                        | 5%          | WB     | -11.0 | -0.9  | 24.9  | 38.6  | 52.7  | 63.9  | 68.4  | 66.0  | 54.3  | 36.0  | 14.5  | -9.7  |  |  |
|                                                                        |             | MCDB   | -10.5 | 0.1   | 29.5  | 48.1  | 65.8  | 75.5  | 79.0  | 75.4  | 60.0  | 39.6  | 15.5  | -9.3  |  |  |
|                                                                        | 10%         | WB     | -14.6 | -5.4  | 20.5  | 35.9  | 50.1  | 61.9  | 66.7  | 63.8  | 51.7  | 32.9  | 9.7   | -14.4 |  |  |
|                                                                        |             | MCDB   | -14.3 | -4.6  | 24.4  | 43.6  | 61.5  | 73.1  | 76.8  | 71.6  | 56.1  | 35.7  | 10.8  | -14.0 |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 14.2  | 23.1  | 31.7  | 28.5  | 26.3  | 28.4  | 25.1  | 24.2  | 20.5  | 16.9  | 15.4  | 11.3  |  |  |
|                                                                        | 5% DB       | MCDBR  | 15.3  | 25.7  | 33.8  | 31.9  | 38.1  | 38.9  | 35.1  | 33.7  | 28.3  | 18.0  | 15.8  | 12.6  |  |  |
|                                                                        |             | MCWBR  | 14.9  | 24.3  | 27.6  | 19.5  | 20.4  | 18.9  | 17.4  | 19.4  | 17.7  | 12.3  | 14.3  | 12.3  |  |  |
|                                                                        | 5% WB       | MCDBR  | 15.2  | 25.4  | 33.2  | 30.4  | 34.1  | 31.2  | 28.8  | 29.6  | 24.5  | 17.2  | 15.8  | 12.8  |  |  |
|                                                                        |             | MCWBR  | 14.8  | 24.0  | 27.1  | 19.0  | 18.5  | 17.1  | 15.9  | 18.0  | 16.1  | 12.1  | 14.3  | 12.5  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.241 | 0.278 | 0.319 | 0.402 | 0.462 | 0.444 | 0.484 | 0.417 | 0.347 | 0.310 | 0.254 | 0.226 |  |  |
|                                                                        | taud        |        | 2.042 | 2.048 | 2.004 | 1.886 | 1.980 | 2.186 | 2.090 | 2.280 | 2.409 | 2.220 | 2.045 | 1.981 |  |  |
|                                                                        | Ebn at Noon |        | 190   | 236   | 257   | 251   | 244   | 252   | 238   | 248   | 252   | 222   | 179   | 154   |  |  |
|                                                                        | Edh at Noon |        | 17    | 28    | 39    | 53    | 51    | 43    | 46    | 36    | 27    | 23    | 17    | 13    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 227   | 565   | 1090  | 1633  | 1585  | 1761  | 1631  | 1294  | 828   | 476   | 290   | 149   |  |  |
|                                                                        | RadStd      |        | 14    | 30    | 41    | 107   | 130   | 122   | 119   | 111   | 65    | 50    | 16    | 5     |  |  |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (2 neighbors) | +0.90 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page